

- 4 (a) Given that $y = (x-1)\sqrt{4x+1}$, show that $\frac{dy}{dx}$ can be written in the form $\frac{px+q}{\sqrt{4x+1}}$ where p and q are constants. [4]

$$\text{Let } u = x-1$$

$$\frac{du}{dx} = 1$$

$$v = \sqrt{4x+1}$$

$$v = (4x+1)^{\frac{1}{2}}$$

$$\frac{dv}{dx} = \frac{1}{2}(4x+1)^{-\frac{1}{2}}(4) = \frac{2}{\sqrt{4x+1}}$$

$$\therefore \frac{dy}{dx} = \sqrt{4x+1} + \frac{2(x-1)}{\sqrt{4x+1}}$$

$$= \frac{4x+1 + 2x-2}{\sqrt{4x+1}}$$

$$= \frac{6x-1}{\sqrt{4x+1}} \quad (\text{shown})$$

- (b) Hence evaluate $\int_2^6 \frac{x}{\sqrt{4x+1}} dx$ [5]

$$\int_2^6 \frac{6x-1}{\sqrt{4x+1}} dx = \left[(x-1)\sqrt{4x+1} \right]_2^6$$

$$\int_2^6 \frac{6x}{\sqrt{4x+1}} dx - \int_2^6 \frac{1}{\sqrt{4x+1}} dx = \left[(6-1)\sqrt{24+1} \right] - \left[(2-1)\sqrt{8+1} \right]$$

$$6 \int_2^6 \frac{x}{\sqrt{4x+1}} dx = \int_2^6 (4x+1)^{-\frac{1}{2}} dx + 5(5) - (1)(3)$$

$$= \left[\frac{[4x+1]^{-\frac{1}{2}+1}}{\frac{1}{2}(4)^{\frac{1}{2}}} \right]_2^6 + 25 - 3$$

$$= \frac{1}{2} \left[\sqrt{4x+1} \right]_2^6 + 22$$

$$= \frac{1}{2} \left[\sqrt{25} - \sqrt{9} \right] + 22$$

$$= \frac{1}{2} [5-3] + 22$$

$$= 23$$

$$\therefore \int_2^6 \frac{x}{\sqrt{4x+1}} dx = \frac{23}{6}$$